ANDRES IGNACIO CARDENAS OYARZUN

Process System Engineer

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Education

Carnegie Mellon University *Pittsburgh, USA*

Ph.D. in Chemical Engineering, GPA: 4.0 (PI: Prof. Ana I. Torres) *2024 – Present*

- Ph.D. Topic: Game-theoretical and Reinforcement Learning frameworks for Chemical Process Design.
- Selected coursework: Advanced Process Systems Engineering (Prof. Ignacio Grossmann), Large-Scale Mathematical Methods (Prof. Lorenz T. Biegler), and Advanced Topics in Machine Learning and Game Theory (Prof. Fei Fang).

Faculty of Physical and Mathematical Sciences, University of Chile *Santiago, Chile*

M.Sc. in Chemical Engineering, GPA: 4.0 *(Valedictorian)* *2022 – 2023*

- Focus: Renewable Energy, Hydrogen, and Process Systems Engineering for sustainability.
- Recipient of the "Best Chemical Engineering Master's Thesis" of the year award.

Faculty of Physical and Mathematical Sciences, University of Chile *Santiago, Chile*

Bachelor's and Professional Degree in Chemical Engineering, GPA: 4.0 *(Valedictorian)* *2017 – 2023*

- Selected coursework: Sustainable Process Optimization, Renewable Energy & Biomass Systems, and Advanced Mass Transfer.
- Participated in the Beauchef Student Orchestra board, exercising a role of leadership.

Experience

University of Chile Energy Center *Santiago, Chile*

Associated researcher *– "Hydrogen Strategies for Chile" (Energy Ministry of Chile)* *01/2023 – 12/2023*

Policy Development for the Deployment of Nation-Wide Hydrogen Systems:

- **Multi-Disciplinary Research Leadership:** Team leader in designing and engineering an evaluation and investment framework for large-scale hydrogen production in Chile.
- **Effectiveness in Industrial Investment:** Led design and implementation of guidelines for the Energy Ministry to incentivize investments for large-scale hydrogen systems nationwide.
- **Strategic Implementation of Public Instruments:** Created a decision-making framework to align the instruments and legal mandates of different departments for the deployment of Hydrogen technologies and infrastructure.

Green Hydrogen – Impacts and Responsibilities in Land Use

- **Competitive Land Disposal for Hydrogen Initiatives:** Directed development of a mathematical model to rank land suitability for hydrogen production across Chile.
- **Responsible Green Hydrogen:** Led a team for research and development of technical, environmental and socioeconomic criteria to assess the impacts of a Hydrogen initiative and its correspondence to a certain land use.

Celulosa Arauco y Constitución *Licantén, Chile*

Process Engineering Intern *– Process Engineering Department* *01/2022 – 03/2022*

Lime Kiln, Caustification and Water Treatment:

- **Calcium Oxide Feed Control:** Developed a predictive model for calcium oxide feed control based on upstream discrete measurements of concentration, temperature, and flow.
- **Flowsheet design:** Design of in-time process flowsheets for continuous monitoring of the lime kiln, caustification, and water treatment operations.

Vegusta S.P.A *Santiago, Chile*

External process consultant *06/2020 – 06/2021*

Operational optimization:

- **Batch scheduling:** Developed a batch optimization scheduling software for the plant's weekly production, aiming

to reduce storage usage under demand.

- **Convection control system:** Formulated an optimization problem that improves the control system of an intermittent air refrigeration unit, minimizing power consumption under time constraints.

M.O. Consultancy *Santiago, Chile*

Part-time consultant *12/2019 – 02/2020*

Patient scheduling: Developed a patient-testing scheduling model to minimize idle time in a public-health hospital.

Publications

Multi-Agent Chemical Process Design: Game Theory and Reinforcement Learning

Andrés I. Cárdenas, Ana I. Torres. (In preparation)

Strategic Design of CO_2-Reuse Pathways for Sustainable Aviation Fuel: A Game-Theoretic Techno-Economic Analysis

Andrés I. Cárdenas, Víctor Soria-Alcaide, Ana I. Torres. (In preparation)

Spatial suitability for green hydrogen and derivatives: A GIS-AHP framework with export-certification sensitivity applied to Chile

Andrés I. Cárdenas, Felipe Díaz-Alvarado, et al. (Available in International Journal of Hydrogen Energy)

Multi-agent H2-MeOH-NH3 industrial park design: Economic and autonomy optimization with simultaneous carrier selection

Andrés I. Cárdenas, Felipe Díaz-Alvarado, Ana I. Torres (In preparation)

Green Hydrogen Production: Process design and capacity expansion integrating economic and operational autonomy objectives.

Andrés I. Cárdenas, Ana I. Torres, Felipe Díaz-Alvarado (Available in Industrial & Engineering Chemistry Research)

Conference Papers and Presentations

Computer-Aided Process Engineering Forum (CAPE) *2026*

Speaker on the use of distributed Reinforcement Learning for the decentralized design of chemical production networks.

European Symposium on Computer Aided Process Engineering (ESCAPE36) *2026*

Speaker on game-theoretical and mathematical optimization frameworks for multi-agent process design in the production of syngas and sustainable aviation fuel through gasification, reforming, and Fischer-Tropsch technologies.

Green Hydrogen in Academia: Bridging the Gap between the Academic and Private Sector *2023*

Member of the winning Chilean team, proposing Process System Engineering (PSE) as a fundamental tool for shared infrastructure planning.

Skills

***Language**:* English (Fluent), Spanish (Native)

Software: Aspen Plus, GAMS, Python/PYOMO, Julia/JuMP, MATLAB, Excel

Teamwork: Active listening, cross-functional collaboration, multi-disciplinary coordination and technical leadership.

Chemical Engineering: Process simulation and optimization, Reinforcement Learning for process design applications.

Extra-curricular activities

Tartan Salsa: *President of the Tartan Salsa (dance club) student association 2026*

Classical music composition: *Student of pianist/cellist and composer Andrés Maupoint/Sebastián Molina (2017 – 2022)*

Orchestra board member: *Founder and directive board member of the Beauchef Student Orchestra (2017 – 2019)*